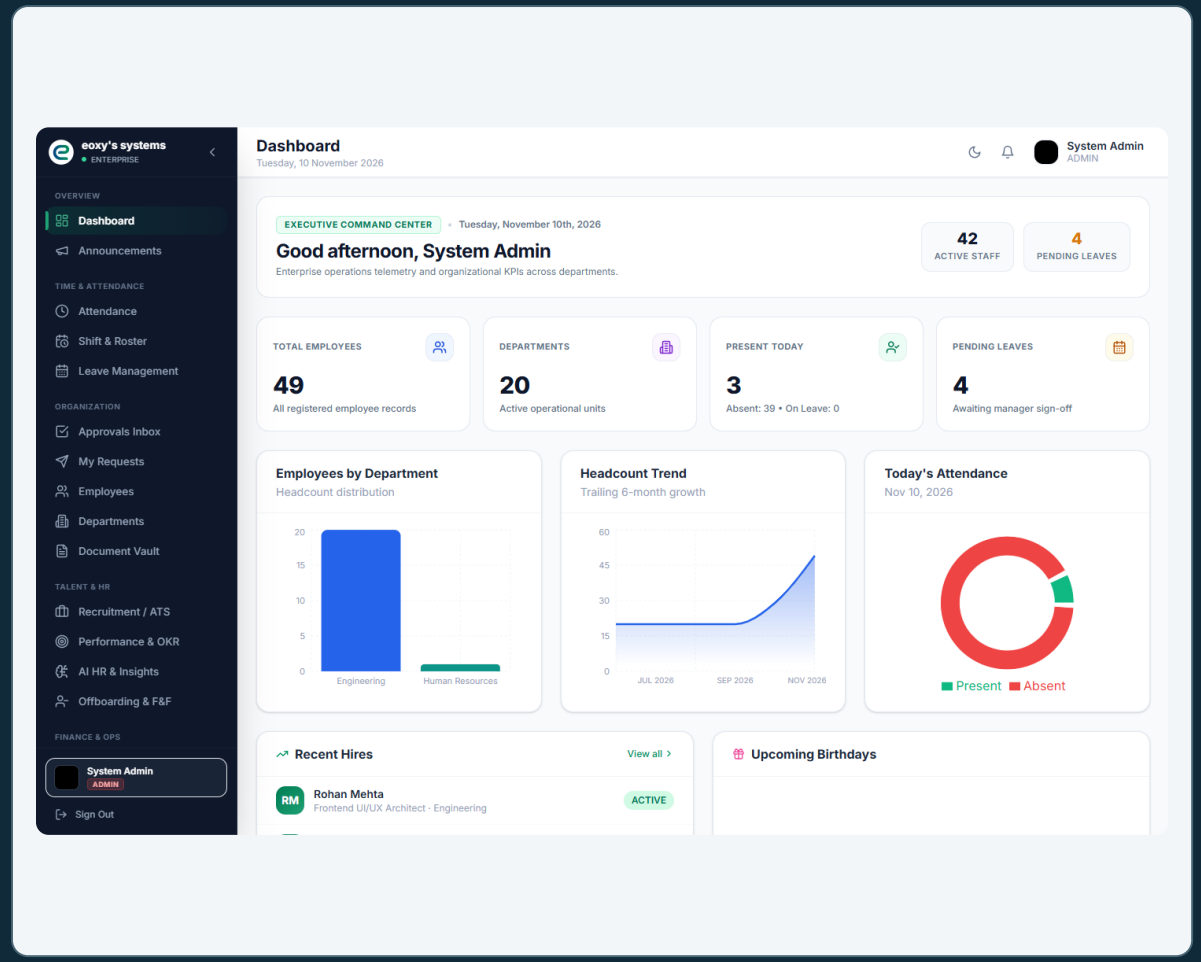
HRMS PLATFORM

Project Analysis and Product Overview

A source-based walkthrough of the product, architecture, modules, user journeys, and implementation findings.

Employee experience · HR operations · Tenant administration



Repository UI capture. Dashboard metrics shown are sample screen data.

11 November 2026

PRODUCT SCOPE

A broad HR operations platform

Implemented feature areas

PEOPLE

- Employees and departments
- Profile and documents
- Recruitment and interviews
- Performance goals and reviews

WORKDAY

- Attendance and policies
- Leave and balances
- Shift rosters and swaps
- Employee requests and approvals

OPERATIONS

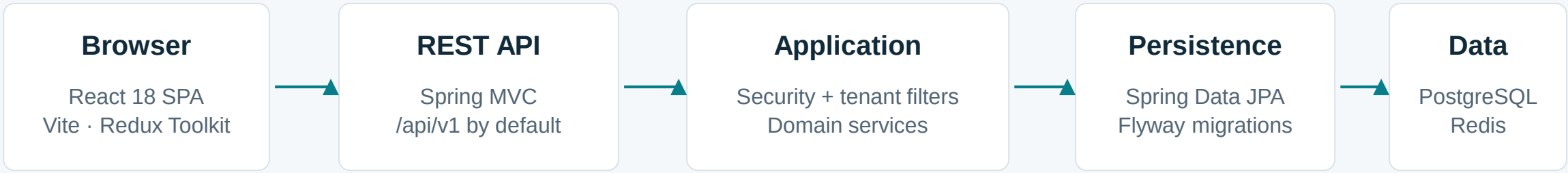
- Payroll and expenses
- Assets and offboarding
- Helpdesk and announcements
- Reports and workforce intelligence

The interface and API are delivered as one multi-tenant application. Module presence in source does not establish that every workflow is production-complete.

SYSTEM ARCHITECTURE

A modular monolith behind a React SPA

One web client calls a Spring Boot API. Services coordinate domain rules and persistence.



External boundaries in code: SMTP email, local upload storage, Redis token blacklist, Prometheus metrics, and interview token/signaling components.

No confirmed live bank or biometric-vendor client was found. Biometric inputs are exposed through REST and CSV ingestion paths.

Backend modules share a common data layer

Java 21 target · Spring Boot 3.2.3

- Controllers validate requests and define REST routes.
- Services coordinate business operations across modules.
- Repositories and JPA entities persist domain records.
- Flyway scripts V1–V15 evolve the relational schema.

Tenant-aware relational model

Core records include users, employees, departments, attendance, leave, payroll, notifications, audit history, and tenant configuration.

Domain tables extend into workflow approvals, documents, expenses, interviews, shifts, assets, offboarding, and workforce intelligence.

Review item: V1 contains MySQL-style index/column alteration syntax while the current runtime configuration targets PostgreSQL.

Access depends on identity, role, and tenant

Visible application roles

Employee

Manager

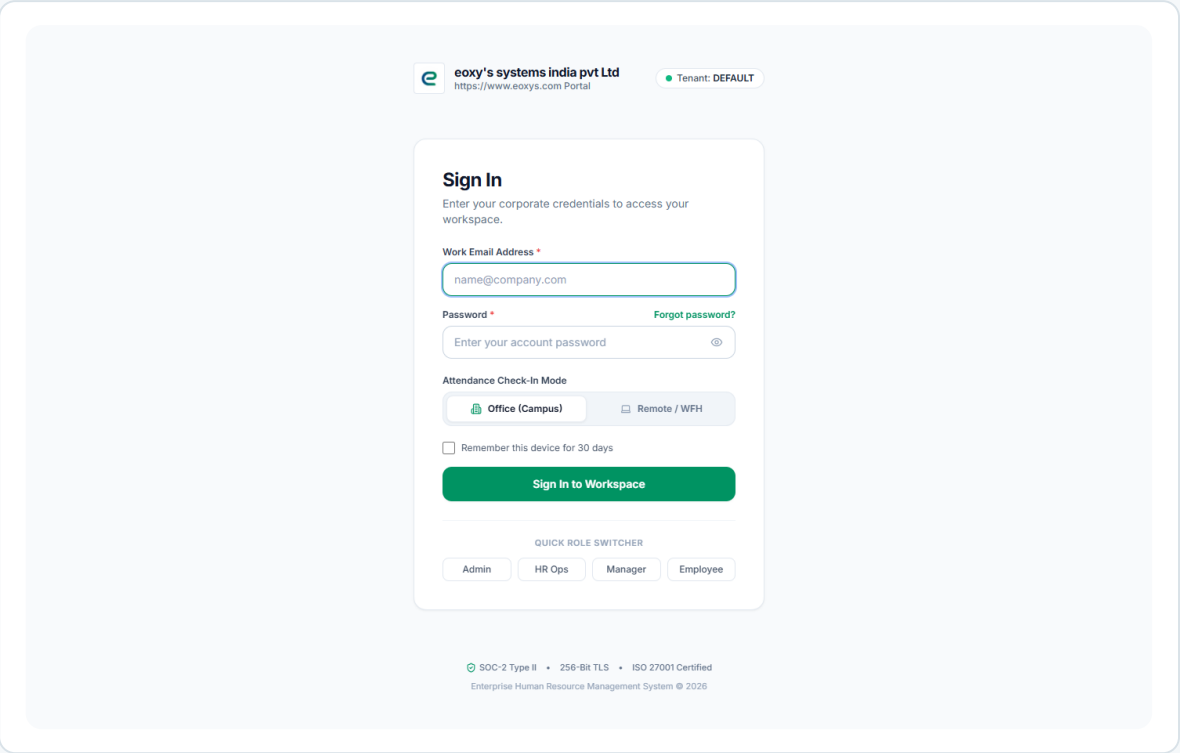
HR

Tenant Admin

Super Admin

IT Admin / IT Support

The app supports JWT authentication, email verification, password reset, first-login change, and TOTP setup endpoints.



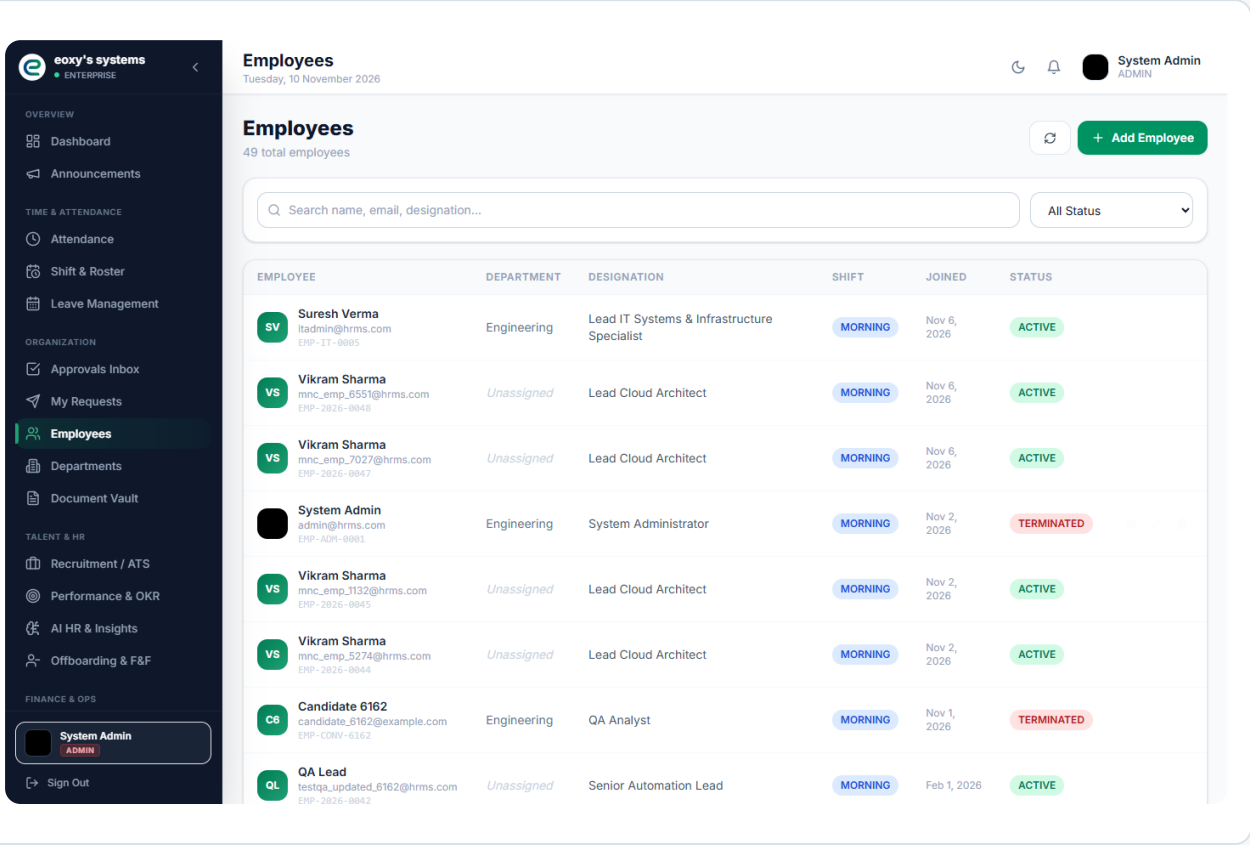
Sign-in UI from the repository capture set

Employee records anchor daily HR operations

Employee directory and profile management

- Search and browse employee records.
- Maintain department and profile details.
- Upload profile photos and employee documents.
- Connect people data to attendance, leave, payroll, and offboarding.

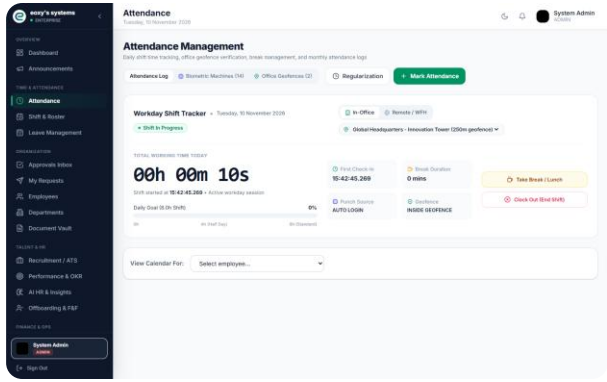
EmployeeController and EmployeeService provide the backend record paths.



Attendance and leave are connected to approval paths

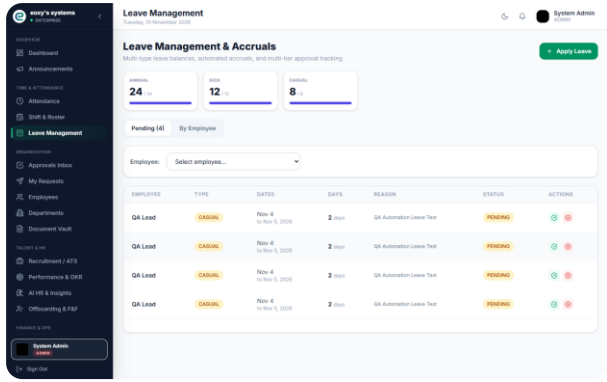
Attendance

Punches, breaks, biometric/CSV ingestion, geofence checks, regularization, office locations, and configurable policies.



Leave

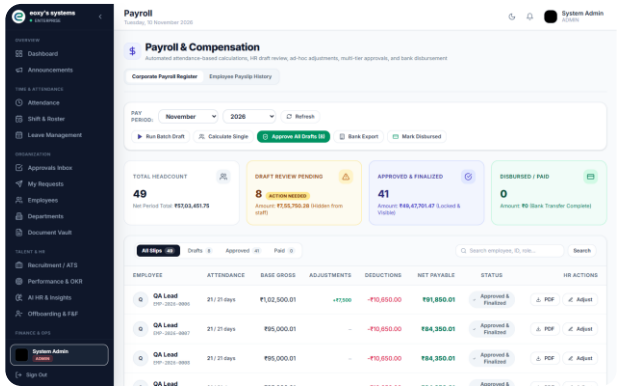
Applications, policy configuration, balance ledgers, accrual scheduling, review, cancellation, and employee notifications.



Payroll and expenses share financial workflows

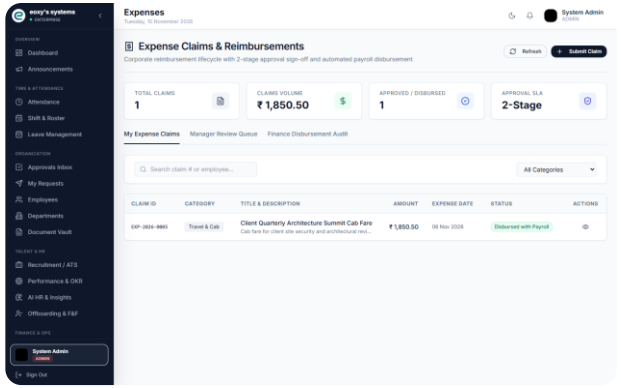
Payroll operations

Process, adjust, approve, mark paid, view registers and summaries, and export bank batch data.



Expense claims

Submit claims and expose manager and finance review steps. Payroll integration should be verified from the service transaction.



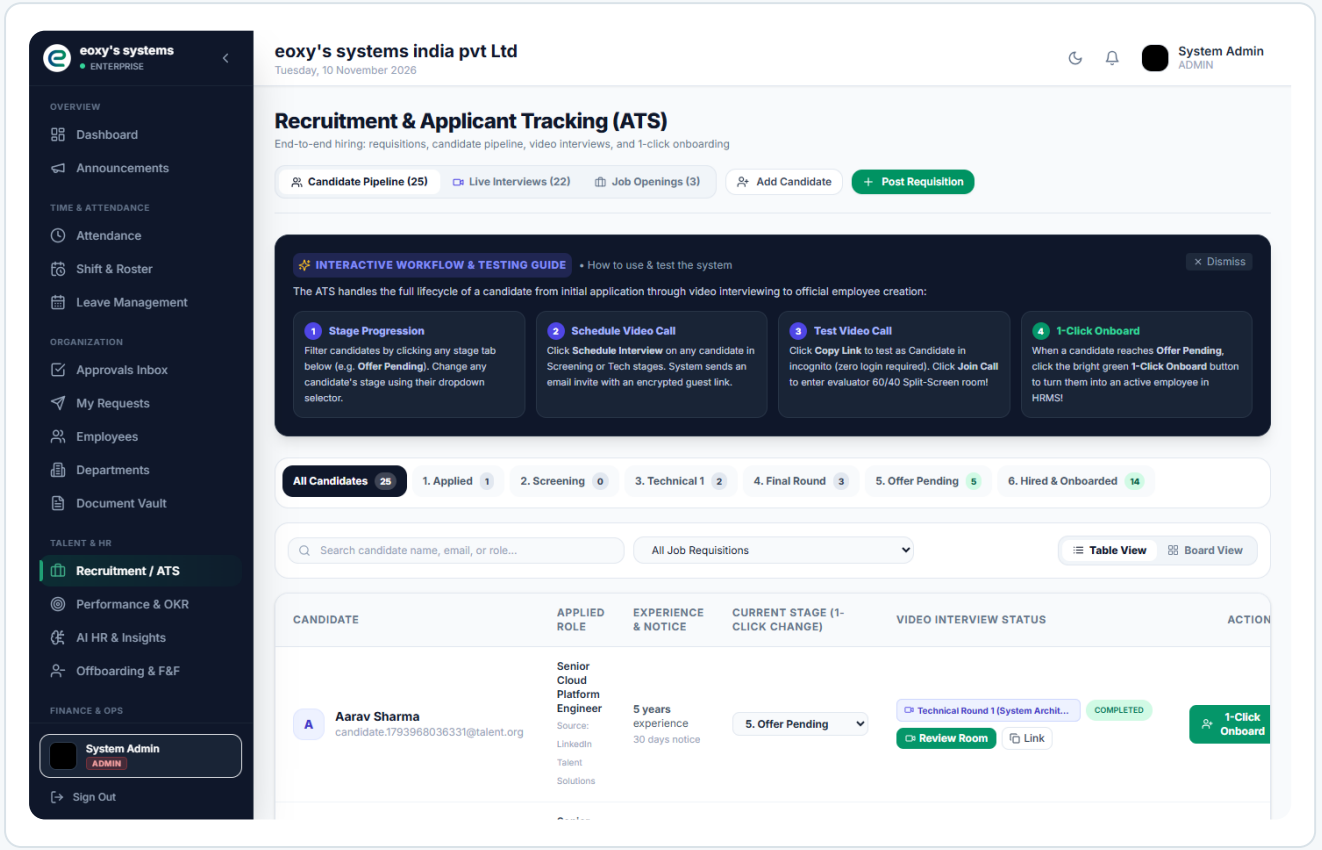
Payroll formulas and statutory settings require payroll-owner confirmation.

Recruitment extends into live interview sessions

Candidate lifecycle surfaces

- Manage jobs and candidate records.
- Update applicant stages and scorecards.
- Schedule interview sessions and panelists.
- Expose candidate guest join and signaling routes.
- Convert candidate records into employee onboarding.

Verify media provider configuration and conversion atomicity before relying on this flow.



Workflows coordinate employee requests

Workflow engine

Workflow definitions contain ordered approval steps. Instances track decisions and status. Leave, attendance regularization, and employee requests have completion handling.

Supporting services

Helpdesk tickets, announcements, document acknowledgements, shifts, assets, performance, offboarding, reports, and notifications have API and UI surfaces.

Example request lifecycle

01 Submit a request

02 Resolve active workflow

03 Reviewer records decision

04 Update domain status

05 Notify requester

Container and monitoring assets support deployment

Technology foundations

Frontend: React 18, Vite, TypeScript/JavaScript, Redux Toolkit, Axios, React Router, Tailwind.

Backend: Java 21 target, Spring Boot 3.2.3, Spring Security, JPA, Flyway, PostgreSQL, Redis, SMTP.

Repository operations assets

- Docker Compose for PostgreSQL, Redis, backend, frontend, Prometheus, Grafana.
- Nginx and Vercel deployment configuration.
- Actuator health and Prometheus metrics.
- JUnit/Spring and Playwright tests plus scripted E2E checks.

Existing tests were inventoried but not executed during this documentation task. Deployment profiles and CI status need environment confirmation.

Production readiness needs a few focused confirmations

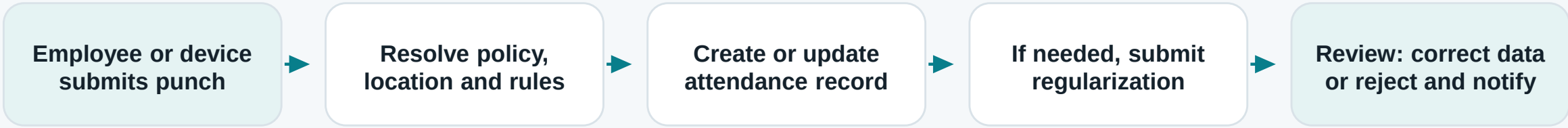
Database	V1 SQL syntax appears MySQL-oriented; reconcile with PostgreSQL migration history.
Session security	logout-all has no visible revocation logic; refresh cookie uses secure(false).
Tenant boundaries	Confirm tenant and record ownership checks on ID-based self-service endpoints.
Business rules	Validate payroll, offboarding and expense-to-payroll behavior with domain owners.
External services	Confirm live interview/media setup, file persistence, and Redis fallback behavior.

Evidence is drawn from the checked-out source tree. Treat these items as review questions, not confirmed production incidents.

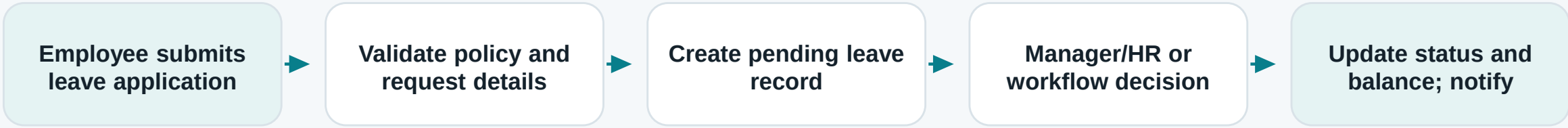
MODULE FLOWCHARTS

Core HR flows connect records to decisions

ATTENDANCE AND REGULARIZATION



LEAVE REQUEST AND APPROVAL

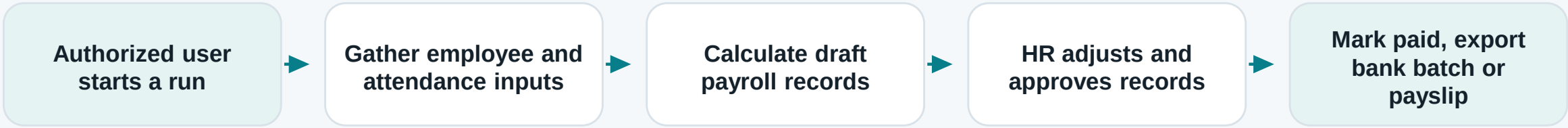


Attendance records feed reporting and payroll inputs. Approved leave updates a matching balance when found; workflow behavior depends on tenant configuration.

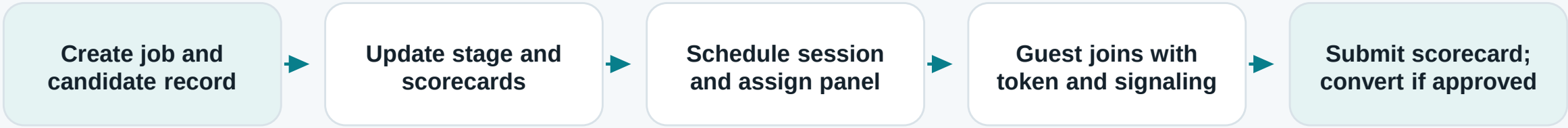
MODULE FLOWCHARTS

Payroll and hiring have distinct review paths

PAYROLL PROCESSING



RECRUITMENT AND INTERVIEW



Payroll formulas, statutory behavior and bank acceptance need owner confirmation. Candidate conversion atomicity and external interview media readiness need review.